



demo7-data-mongo: Files required

1. MyBootApplication.java	Updated in Demo7
2. Customer.java	Same as Demo6
3. CustomerDAO.java	Updated in Demo7
4. application.properties	Same as Demo6
5. pom.xml	Same as Demo6

1) MyBootApplication.java

```
package com.jlcindia.springboot;

import java.util.List;

import org.springframework.beans.factory.annotation.Autowired;
import org.springframework.boot.CommandLineRunner;
import org.springframework.boot.SpringApplication;
import org.springframework.boot.autoconfigure.SpringBootApplication;
/*
 * @Author : Srinivas Dande
 * @Company: Java Learning Center
 */

@SpringBootApplication
public class MyBootApplication implements CommandLineRunner {

    @Autowired
    CustomerDAO custDAO;

    @Override
    public void run(String... args) throws Exception {

        List<Customer> mylist=null;

        //1. getCustomersByCname
        mylist=custDAO.getCustomersByCname("Sri");
        mylist.forEach(System.out::println);
        System.out.println("-----");

        //2. getCustomersByEmail
        mylist=custDAO.getCustomersByEmail("hello@jlc");
        mylist.forEach(System.out::println);
        System.out.println("-----");
    }
}
```



//3. readCustomersByPhone

```
mylist=custDAO.readCustomersByPhone(12345);  
mylist.forEach(System.out::println);  
System.out.println("-----");
```

//4. findCustomersByCity

```
mylist=custDAO.findCustomersByCity("Hyd");  
mylist.forEach(System.out::println);  
System.out.println("-----");
```

//5. getCustomersByEmailAndPhone

```
mylist=custDAO.getCustomersByEmailAndPhone("hello@jlc", 12345);  
mylist.forEach(System.out::println);  
System.out.println("-----");
```

//6. getCustomersByEmailOrPhone

```
mylist=custDAO.getCustomersByEmailOrPhone("hello@jlc", 12345);  
mylist.forEach(System.out::println);  
System.out.println("-----");
```

//7. getCustomersByCityIn

```
mylist=custDAO.getCustomersByCityIn("Hyd","Delhi");  
mylist.forEach(System.out::println);  
System.out.println("-----");
```

//8. getCustomersByCityNotIn

```
mylist=custDAO.getCustomersByCityNotIn("Hyd","Delhi");  
mylist.forEach(System.out::println);
```

//9. countCustomersByCity

```
long count=custDAO.countCustomersByCity("Hyd");  
System.out.println(count);  
System.out.println("-----");
```

//10. countCustomersByCityIn

```
count=custDAO.countCustomersByCityIn("Hyd","Delhi");  
System.out.println(count);  
System.out.println("-----");
```

//11. getCustomersByCityNotIn

```
count=custDAO.countCustomersByCityNotIn("Blore");  
System.out.println(count);  
System.out.println("-----");
```



```
//12. getCustomersByBalance
mylist=custDAO.getCustomersByBalance(15000);
mylist.forEach(System.out::println);
System.out.println("-----");

//13. getCustomersByBalanceIsBetween
mylist=custDAO.getCustomersByBalanceIsBetween(10000,20000);
mylist.forEach(System.out::println);
System.out.println("-----");

//14. getCustomersByBalanceIsGreaterThan
mylist=custDAO.getCustomersByBalanceIsGreaterThan(20000);
mylist.forEach(System.out::println);
System.out.println("-----");

//15. getCustomersByBalanceLessThan
mylist=custDAO.getCustomersByBalanceLessThan(10000);
mylist.forEach(System.out::println);
System.out.println("-----");

//16. getCustomersByBalanceGreaterThanOrEqualTo
mylist=custDAO.getCustomersByBalanceGreaterThanOrEqualTo(20000);
mylist.forEach(System.out::println);
System.out.println("-----");

//17. getCustomersByBalanceLessThanOrEqualTo
mylist=custDAO.getCustomersByBalanceLessThanOrEqualTo(10000);
mylist.forEach(System.out::println);
System.out.println("-----");

System.out.println("Done!!!");

}
public static void main(String[] args) {
    SpringApplication.run(MyBootApplication.class, args);
}
}
```



3) CustomerDAO.java

```
package com.jlcindia.springboot;

import java.util.List;

import org.springframework.data.mongodb.repository.MongoRepository;
import org.springframework.stereotype.Repository;
/*
 * @Author : Srinivas Dande
 * @Company: Java Learning Center
 */
@Repository
public interface CustomerDAO extends MongoRepository<Customer, String> {

    List<Customer> getCustomersByCname(String cname);
    List<Customer> getCustomersByEmail(String email);
    List<Customer> readCustomersByPhone(int phone);
    List<Customer> findCustomersByCity(String city);

    List<Customer> getCustomersByEmailAndPhone(String email, int phone);
    List<Customer> getCustomersByEmailOrPhone(String email, int phone);

    List<Customer> getCustomersByCityIn(String... cities);
    List<Customer> getCustomersByCityNotIn(String... cities);

    long countCustomersByCity(String city);
    long countCustomersByCityIn(String... cities);
    long countCustomersByCityNotIn(String... cities);

    List<Customer> getCustomersByBalance(double bal);
    List<Customer> getCustomersByBalanceIsBetween(double bal1, double bal2);

    List<Customer> getCustomersByBalanceIsGreaterThan(double bal);
    List<Customer> getCustomersByBalanceLessThan(double bal);

    List<Customer> getCustomersByBalanceGreaterThanEqual(double bal);
    List<Customer> getCustomersByBalanceLessThanEqual(double bal);
}
```



demo8-data-mongo: Files required

1. MyBootApplication.java	Updated in Demo8
2. Customer.java	Same as Demo7
3. CustomerDAO.java	Updated in Demo8
4. application.properties	Same as Demo7
5. pom.xml	Same as Demo7

1) MyBootApplication.java

```
package com.jlcindia.springboot;

import java.util.List;

import org.springframework.beans.factory.annotation.Autowired;
import org.springframework.boot.CommandLineRunner;
import org.springframework.boot.SpringApplication;
import org.springframework.boot.autoconfigure.SpringBootApplication;
/*
 * @Author : Srinivas Dande
 * @Company: Java Learning Center
 */
@SpringBootApplication
public class MyBootApplication implements CommandLineRunner {

    @Autowired
    CustomerDAO custDAO;

    @Override
    public void run(String... args) throws Exception {

        List<Customer> mylist=null;

        //1. fetchCustomersByCity
        mylist=custDAO.fetchCustomersByCity("Hyd");
        mylist.forEach(System.out::println);
        System.out.println("-----");

        //2. fetchCustomersByPhone
        mylist=custDAO.fetchCustomersByPhone(12345);
        mylist.forEach(System.out::println);

        //3. fetchCustomersByBal
        mylist=custDAO.fetchCustomersByBal(15000);
        mylist.forEach(System.out::println);
        System.out.println("-----");
    }
}
```



```
//4. fetchCustomersByBalGTE
mylist=custDAO.fetchCustomersByBalGTE(15000);
mylist.forEach(System.out::println);
System.out.println("-----");

//5. fetchCustomersByBalLTE
mylist=custDAO.fetchCustomersByBalLTE(15000);
mylist.forEach(System.out::println);
System.out.println("-----");

//6. fetchCustomersByBalRange
mylist=custDAO.fetchCustomersByBalRange(10000,20000);
mylist.forEach(System.out::println);
System.out.println("-----");

//7. fetchCustomersByEmailAndPhone
mylist=custDAO.fetchCustomersByEmailAndPhone("hello@jlc",12345);
mylist.forEach(System.out::println);
System.out.println("-----");

//8. fetchCustomersByEmailOrPhone
mylist=custDAO.fetchCustomersByEmailOrPhone("hello@jlc",12345);
mylist.forEach(System.out::println);
System.out.println("-----");

//9. fetchCustomersByCitiesIN
mylist=custDAO.fetchCustomersByCitiesIN("Hyd","Delhi");
mylist.forEach(System.out::println);
System.out.println("-----");

//10. fetchCustomersByCitiesNIN
mylist=custDAO.fetchCustomersByCitiesNIN("Hyd","Delhi");
mylist.forEach(System.out::println);
System.out.println("-----");

System.out.println("Done!!!");

}

public static void main(String[] args) {
    SpringApplication.run(MyBootApplication.class, args);
}

}
```



3) CustomerDAO.java

```
package com.jlcindia.springboot;

import java.util.List;

import org.springframework.data.mongodb.repository.MongoRepository;
import org.springframework.data.mongodb.repository.Query;
import org.springframework.stereotype.Repository;
/*
 * @Author : Srinivas Dande
 * @Company: Java Learning Center
 */
@Repository
public interface CustomerDAO extends MongoRepository<Customer, String> {

    @Query("{ 'city':?0}")
    List<Customer> fetchCustomersByCity(String mycity);

    @Query("{ 'phone':?0}")
    List<Customer> fetchCustomersByPhone(int myphone);

    @Query("{ 'balance':?0}")
    List<Customer> fetchCustomersByBal(double bal);

    @Query("{ 'balance': {$gte:?0}}")
    List<Customer> fetchCustomersByBalGTE(double bal);

    @Query("{ 'balance': {$lte:?0}}")
    List<Customer> fetchCustomersByBalLTE(double bal);

    @Query("{ $and : [ { 'balance': {$gte:?0}} , { 'balance': {$lte:?1}} ]}")
    List<Customer> fetchCustomersByBalRange(double bal1, double bal2);

    @Query("{ $and : [ { 'email': ?0} , { 'phone':?1} ]}")
    List<Customer> fetchCustomersByEmailAndPhone(String email, int phone);

    @Query("{ $or : [ { 'email': ?0} , { 'phone':?1} ]}")
    List<Customer> fetchCustomersByEmailOrPhone(String email, int phone);

    @Query("{ 'city':{$in:?0}}")
    List<Customer> fetchCustomersByCitiesIN(String... mycities);

    @Query("{ 'city':{$nin:?0}}")
    List<Customer> fetchCustomersByCitiesNIN(String... mycities);
}
```



demo9-data-mongo: Files required

1. MyBootApplication.java	Updated in Demo9
2. Customer.java	Same as Demo8
3. CustomerDAO.java	Updated in Demo9
4. application.properties	Same as Demo8
5. pom.xml	Same as Demo8

1) MyBootApplication.java

```
package com.jlcindia.springboot;

import java.util.List;

import org.springframework.beans.factory.annotation.Autowired;
import org.springframework.boot.CommandLineRunner;
import org.springframework.boot.SpringApplication;
import org.springframework.boot.autoconfigure.SpringBootApplication;
import org.springframework.data.domain.Page;
import org.springframework.data.domain.PageRequest;
import org.springframework.data.domain.Pageable;
import org.springframework.data.domain.Sort;
import org.springframework.data.domain.Sort.Order;
/*
 * @Author : Srinivas Dande
 * @Company: Java Learning Center
 */
@SpringBootApplication
public class MyBootApplication implements CommandLineRunner {

    @Autowired
    CustomerDAO custDAO;

    @Override
    public void run(String... args) throws Exception {

        List<Customer> mylist=null;

        //1. findAll - phoneSort
        Sort phoneSort1 = Sort.by(Order.desc("phone"));

        mylist=custDAO.findAll(phoneSort1);
        mylist.forEach(System.out::println);

        System.out.println("-----");
    }
}
```



//1. findAll - emailSort

```
Sort emailSort1 = Sort.by(Order.asc("email"));
```

```
mylist=custDAO. findAll(emailSort1);  
mylist.forEach(System.out::println);  
System.out.println("-----");
```

//2. findAll - mypage1

```
Page<Customer> pageList=null;  
Pageable mypage1= PageRequest.of(0, 5);  
pageList=custDAO. findAll(mypage1);
```

```
pageList.forEach(System.out::println);  
System.out.println("-----");
```

//2. findAll - mypage2

```
Sort emailSort = Sort.by(Order.asc("email"));  
Pageable mypage2= PageRequest.of(0, 5,emailSort);  
pageList=custDAO. findAll(mypage2);  
pageList.forEach(System.out::println);  
System.out.println("-----");
```

//3. fetchCustomersByCity - emailSort

```
mylist=custDAO. fetchCustomersByCity("Blore",emailSort);  
mylist.forEach(System.out::println);
```

//4. fetchCustomersByCity - mypage3

```
List<Customer> custList=null;  
Pageable mypage3= PageRequest.of(0, 3);  
custList=custDAO. fetchCustomersByCity("Blore",mypage3);  
custList.forEach(System.out::println);  
System.out.println("-----");
```

//4. fetchCustomersByCity - mypage4

```
Pageable mypage4= PageRequest.of(0, 3,emailSort);  
custList=custDAO. fetchCustomersByCity("Blore",mypage4);  
custList.forEach(System.out::println);  
System.out.println("-----");
```

```
System.out.println("Done!!!");  
}
```

```
public static void main(String[] args) {  
SpringApplication.run(MyBootApplication.class, args);  
}  
}
```



3) CustomerDAO.java

```
package com.jlcindia.springboot;

import java.util.List;

import org.springframework.data.domain.Pageable;
import org.springframework.data.domain.Sort;
import org.springframework.data.mongodb.repository.MongoRepository;
import org.springframework.data.mongodb.repository.Query;
import org.springframework.stereotype.Repository;
/*
 * @Author : Srinivas Dande
 * @Company: Java Learning Center
 */
@Repository
public interface CustomerDAO extends MongoRepository<Customer, String>{

    @Query("{ 'city' : ?0 }")
    List<Customer> fetchCustomersByCity(String city,Sort sort);

    @Query("{ 'city' : ?0 }")
    List<Customer> fetchCustomersByCity(String city,Pageable pageable);

}
```

demo10-data-mongo: Files required

1. MyBootApplication.java	Updated in Demo10
2. Customer.java	Same as Demo9
3. CustomerDAO.java	Updated in Demo10
4. CustomerDAOImpl.java	Newly Added in Demo10
5. application.properties	Same as Demo9
6. pom.xml	Same as Demo9

1) MyBootApplication.java

```
package com.jlcindia.springboot;

import java.util.List;

import org.springframework.beans.factory.annotation.Autowired;
import org.springframework.boot.CommandLineRunner;
import org.springframework.boot.SpringApplication;
import org.springframework.boot.autoconfigure.SpringBootApplication;
```



```
/*
 * @Author : Srinivas Dande
 * @Company: Java Learning Center
 */
@SpringBootApplication
public class MyBootApplication implements CommandLineRunner {

    @Autowired
    CustomerDAO custDAO;

    @Override
    public void run(String... args) throws Exception {

        // 1.save()
        Customer cust = new Customer("109", "asd", "asd@jlc", 12345, "Blore", 17500);
        custDAO.addCustomer(cust);

        //2. getAllCustomers
        List<Customer> mylist=custDAO.getAllCustomers();
        mylist.forEach(System.out::println);

        System.out.println("Done!!!");
    }

    public static void main(String[] args) {
        SpringApplication.run(MyBootApplication.class, args);
    }
}
```

3) CustomerDAO.java

```
package com.jlcindia.springboot;

import java.util.List;
import org.springframework.stereotype.Repository;

/*
 * @Author : Srinivas Dande
 * @Company: Java Learning Center
 */
@Repository
public interface CustomerDAO {
    public void addCustomer(Customer cust);
    public List<Customer> getAllCustomers();
}
```



4) CustomerDAOImpl.java

```
package com.jlcindia.springboot;

import java.util.List;

import org.springframework.beans.factory.annotation.Autowired;
import org.springframework.data.mongodb.core.MongoTemplate;
import org.springframework.stereotype.Repository;
/*
 * @Author : Srinivas Dande
 * @Company: Java Learning Center
 */
@Repository
public class CustomerDAOImpl implements CustomerDAO {

    @Autowired
    MongoTemplate mongoTemp;

    @Override
    public void addCustomer(Customer cust) {
mongoTemp.save(cust);
    }

    @Override
    public List<Customer> getAllCustomers() {
return mongoTemp.findAll(Customer.class);
    }

}
```